SKILL -8 Spring Boot- JPQL & Query Methods Module

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Skill Experiment Number: 08

Section Number: 75

Code:

Skill8Application.java

```
package com.example.productsearch;


import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;


@SpringBootApplication
public class Skill8Application {


	public static void main(String[] args) {
		SpringApplication.run(Skill8Application.class, args);
	}


}
```

ProductController.java

```
package com.example.productsearch.controller;


import java.util.List;
```

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```java
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;

import com.example.productsearch.entity.Product;
import com.example.productsearch.repository.ProductRepository;

@RestController
@RequestMapping("/products")
public class ProductController {

  @Autowired
  private ProductRepository repo;

  // Add product
  @PostMapping("/add")
  public Product addProduct(@RequestBody Product product) {
    return repo.save(product);
  }

  // Category search
  @GetMapping("/category/{category}")
  public List<Product> getByCategory(@PathVariable String category) {
    return repo.findByCategory(category);
  }

  // Price filter
```

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```
    @GetMapping("/filter")
    public List<Product> getByPriceRange(@RequestParam double min, @RequestParam double max) {
        return repo.findByPriceBetween(min, max);
    }

    // Sorted by price
    @GetMapping("/sorted")
    public List<Product> getSortedProducts() {
        return repo.getProductsSortedByPrice();
    }

    // Expensive products
    @GetMapping("/expensive/{price}")
    public List<Product> getExpensive(@PathVariable double price) {
        return repo.findExpensiveProducts(price);
    }
}
```

Product.java

```
package com.example.productsearch.entity;

import jakarta.persistence.*;

@Entity
public class Product {
```

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```
@Id
@GeneratedValue(strategy = GenerationType.IDENTITY)
private Long id;

private String name;
private String category;
private double price;

public Product() {}

public Product(String name, String category, double price) {
   this.name = name;
   this.category = category;
   this.price = price;
}

// getters and setters
public Long getId() { return id; }

public String getName() { return name; }

public void setName(String name) { this.name = name; }

public String getCategory() { return category; }

public void setCategory(String category) { this.category = category; }
```

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```
    public double getPrice() { return price; }

    public void setPrice(double price) { this.price = price; }
}
```

ProductRepository.java

```
package com.example.productsearch.repository;

import java.util.List;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.data.jpa.repository.Query;

import com.example.productsearch.entity.Product;

public interface ProductRepository extends JpaRepository<Product, Long> {

    // Derived Query Methods
    List<Product> findByCategory(String category);

    List<Product> findByPriceBetween(double min, double max);

    // JPQL Queries

    // Sorting by price
    @Query("SELECT p FROM Product p ORDER BY p.price ASC")
```

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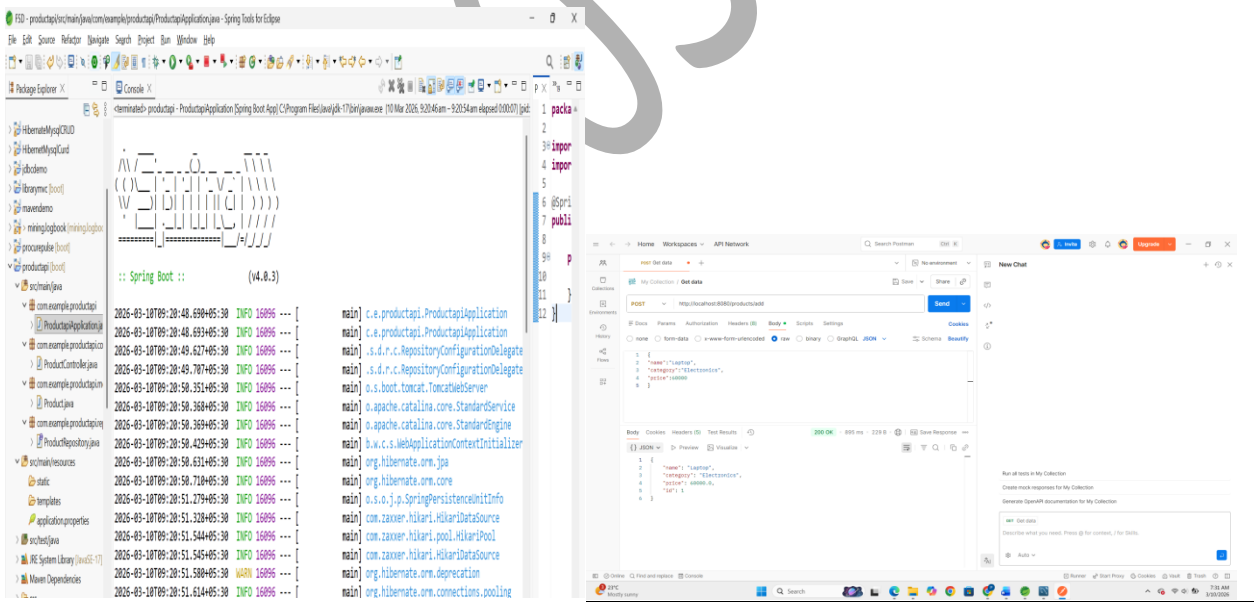
```
    List<Product> getProductsSortedByPrice();


    // Fetch products above price
    @Query("SELECT p FROM Product p WHERE p.price > ?1")
    List<Product> findExpensiveProducts(double price);


    // Fetch by category using JPQL
    @Query("SELECT p FROM Product p WHERE p.category = ?1")
    List<Product> getProductsByCategory(String category);
}
```

Output:



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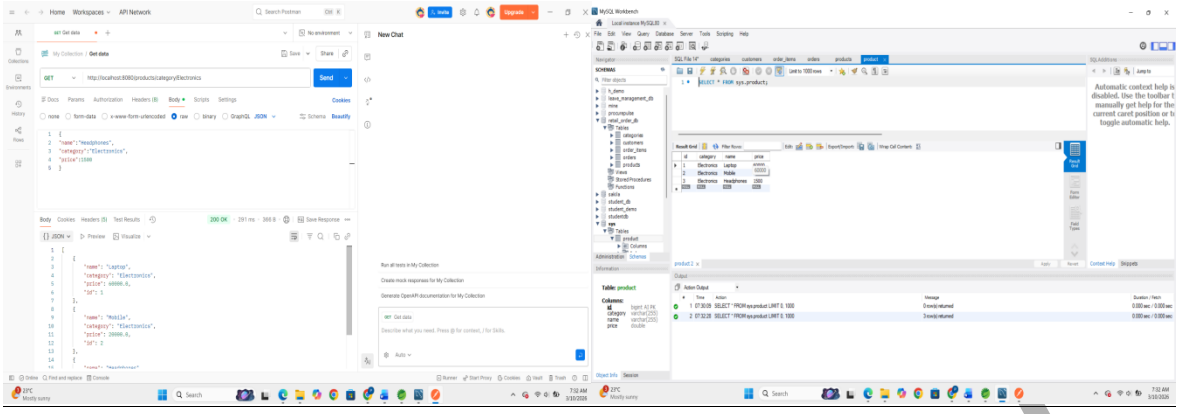

https://github.com/Gowtham-406/S-75_1041_SkillLabDocuments.git

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